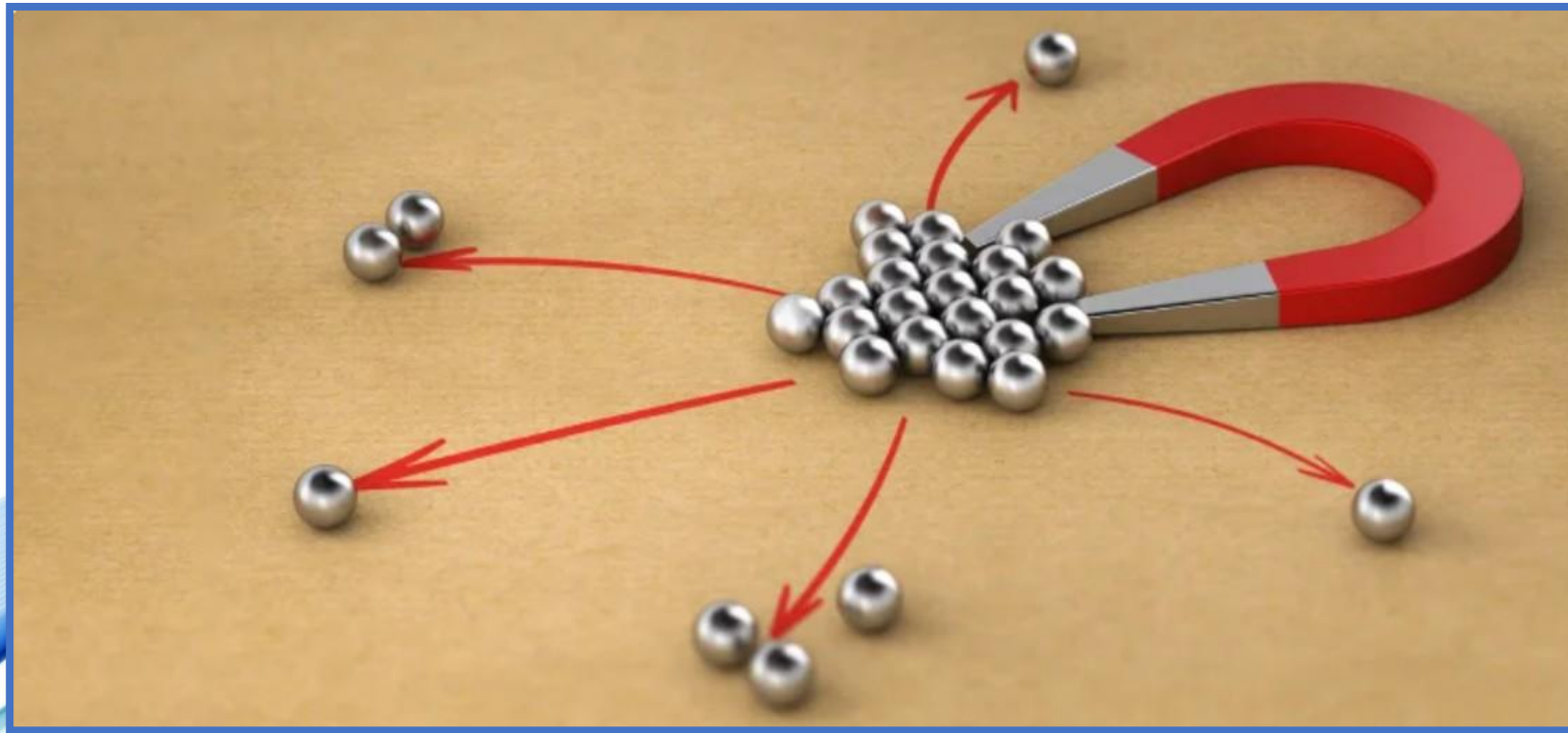


PROJECT CHURNGUARD

Capstone Presentation

By Sivin Stephen, Great Learning, May 26



Business Problem Understanding

The Problem

- Customer attrition is high (**16.84%**).
- Specifically, in **"Regular Plus"** segment churn hits **27%**.

Constraints

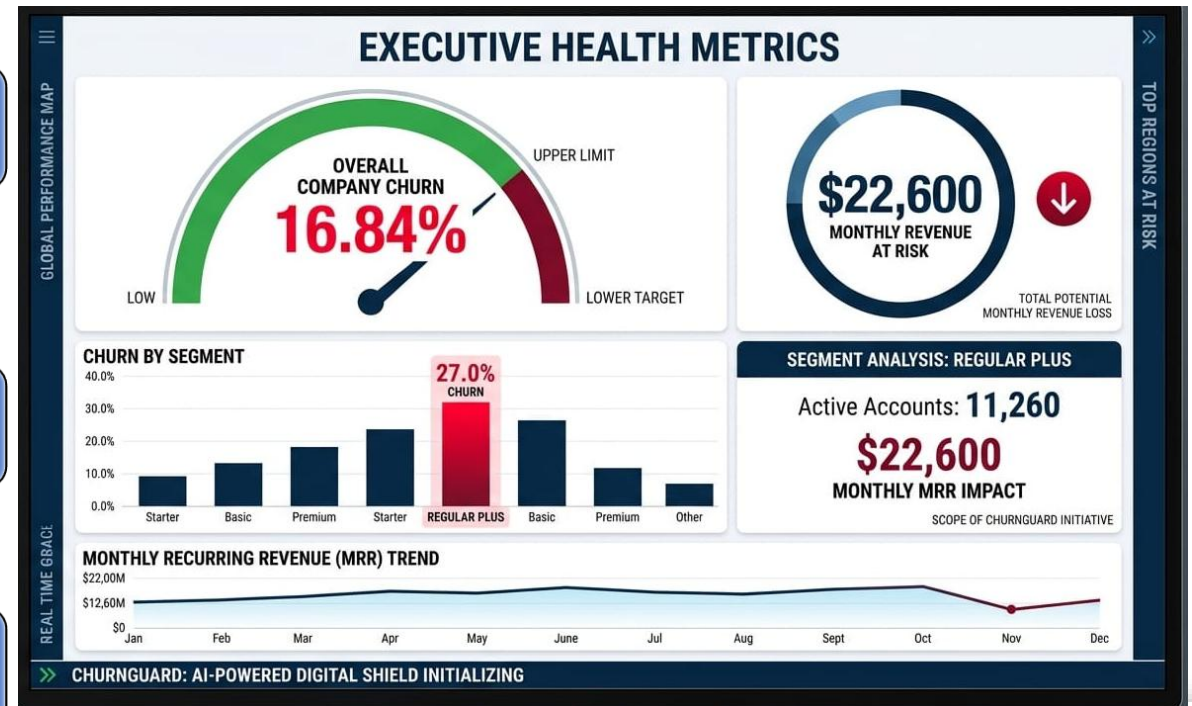
- We **cannot offer discounts to everyone** (expensive); we **need to target** only those truly at risk.

Scope

- **Analysis of 11,260 accounts across** all city tiers and demographics.

Objectives:

- **Identify the "Breaking Point"** where service friction leads to exit.
- **Build a proactive "Early Warning System"** to replace reactive fire-fighting



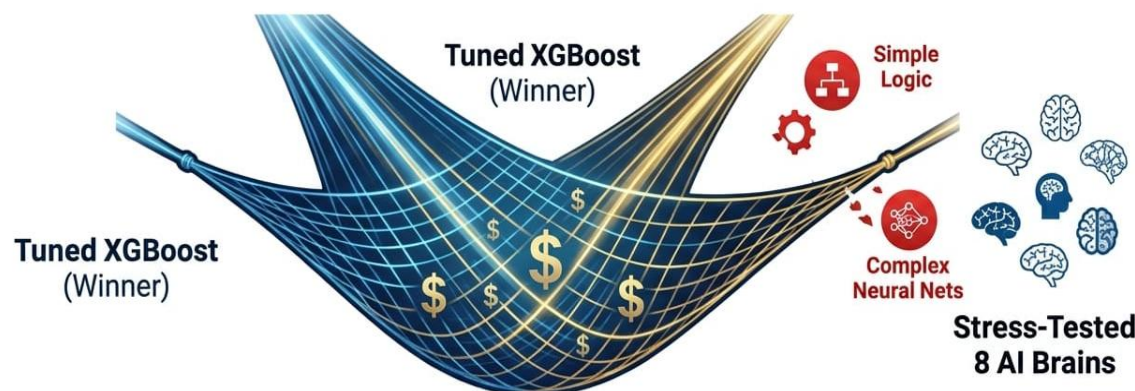
Modelling Approach Used & Why

The Strategy: We tested 8 different AI architectures. We moved away from simple models because customer behavior is complex and "non-linear."

The Winner: Tuned XGBoost Classifier.

Why this model?

- It uses "Gradient Boosting" to **learn from its own mistakes**.
- **High Catch Rate (87% Recall):** It catches 9 out of 10 churners.
- **High Accuracy (94% Precision):** When the model says someone is leaving, it is right 94% of the time.



PRECISION

Surgically Accurate. Won't annoy loyal customers. Flags are right 94% of the time.



CATCH RATE (RECALL)

Identifying 9 out of 10 potential exits before they happen.

This ensures we don't waste retention budgets on happy customers.



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Insights from Analysis

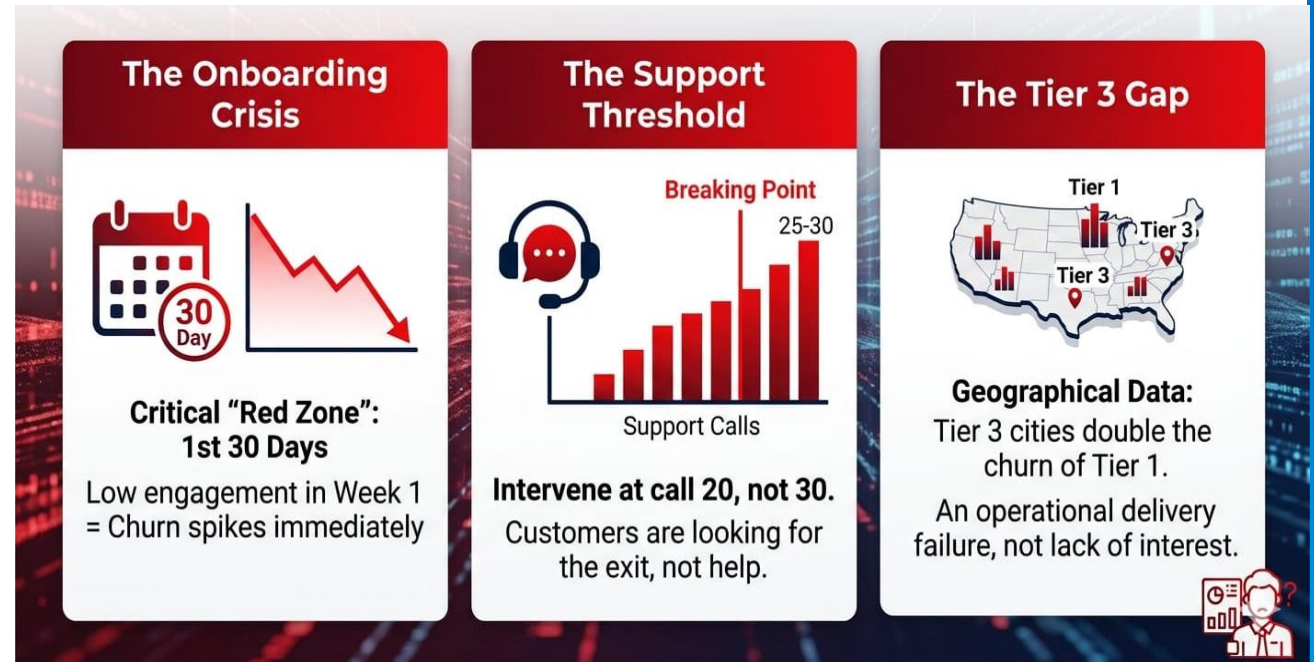
The Onboarding "Red Zone": Most customers leave within the **first 30 days**. If we keep them for 20 months, they stay forever.

The Tier 3 Gap: Customers in Tier 3 cities churn at double the rate of Tier 1, highlighting a localized delivery/infrastructure issue.

Payment Stickiness: Manual payment methods (Cash on Delivery) have 25% higher churn than automated billing.

The Service "Breaking Point":

Churn probability spikes once a customer reaches 25-30 support calls.



Recommendations

The ROI: By saving just half of the predicted churners in the "Regular Plus" segment, the company gains **\$11,000+ per month in recovered revenue.**

✓ **Pillar 1: AI-Powered "Shield":**

Daily database scans to flag high-risk accounts for immediate intervention.

✓ **Pillar 2: Digital Concierge:**

Automated "Welcome Assistant" for new users who show low engagement in Week 1.

✓ **Pillar 3: Tier 3 Service Audit:**

localized infrastructure fix to close the 9% "Geographical Gap."

✓ **Pillar 4: Auto-Pay Incentives:**

Move COD users to Credit/Debit to "lock in" loyalty.

